

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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service representatives, and home health aides by age using data from the CPS. Though there are millions of workers employed in these professions across the US, the estimates are highly volatile, with common fluctuations of 20% or greater in estimated employment month-to-month. Figure A21 shows estimated employment changes by age and exposure quintile using the CPS, also suggesting a high degree of volatility in the estimates.

This volatility in CPS microdata reflects small sample sizes and the fact that the CPS is not stratified to target employment statistics for these demographic-occupation subgroups.²⁷ The sample size and sampling procedure of the CPS may therefore make it challenging to assess employment changes by age and AI exposure with a high degree of confidence over the time horizon considered in this paper (Chandar, 2025b; O’Brien, 2025).

Other large-scale data sources such as the American Community Survey (ACS) may offer a more reliable comparison to the ADP data, though the ACS is released with a significant lag compared to the data from ADP. We encourage comparison of our findings to results from other data sources such as the ACS upon their release.²⁸

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- First, we find substantial declines in employment for early-career workers in occupations most exposed to AI, such as software development and customer support.
- Second, we show that economy-wide employment continues to grow, but employment growth for young workers has been stagnant.
- Third, entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- Fourth, these employment declines remain after conditioning on firm-time effects, with a 13% relative employment decline for young workers in the most exposed occupations.

²⁷The CPS includes between 26 and 53 young software developers aged between 22 and 25 per month over our sample period. It includes between 49 and 95 young customer service representatives, and between 2 and 14 young home health aides.

²⁸The 2024 ACS 1-Year Public Use Microdata Sample is scheduled to be released on October 16, 2025.

- Fifth, these labor market adjustments are more visible in employment than in compensation.
- Sixth, we find that these patterns hold in occupations unaffected by remote work and across various alternative sample constructions.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to significantly affect entry-level employment.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in an adjustment period as workers reallocate from displaced forms of work to new forms with growing labor demand (Autor et al., 2024). Such endogenous adjustment may already be happening with AI, with emerging evidence of shifts in college majors away from AI-exposed categories such as computer science (Horowitz, 2025). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others (Bresnahan et al., 2002; Brynjolfsson et al., 2021).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. Consequently, we will continue to monitor these outcomes to assess whether the trends documented in the paper accelerate in the future. Future work would benefit from better firm-level AI adoption data, which would provide sharper variation for estimating plausible causal effects of AI on employment.

References

- ACEMOGLU, D. AND D. AUTOR (2011): “Skills, Tasks and Technologies: Implications for Employment and Earnings*,” in *Handbook of Labor Economics*, ed. by D. Card and O. Ashenfelter, Elsevier, vol. 4, 1043–1171. 1
- ACEMOGLU, D., D. AUTOR, J. HAZELL, AND P. RESTREPO (2022): “Artificial Intelligence and Jobs: Evidence from Online Vacancies,” *Journal of Labor Economics*, 40, S293–S340, publisher: The University of Chicago Press. 9
- ADP RESEARCH (2025): “ADP National Employment Report,” . 3.1